

A project

report on

“Smart Glove: Translating Gestures into Text”

Submitted for partial fulfillment of the requirements for the

Community Engagement Project (SE, 1nd Semester)

of

Bachelor of Engineering

by

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2025 - 2026

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CERTIFICATE

This is to certify that the project work entitled “Smart Glove: Translating Gestures into Text” is a bona fide work carried out by Ishita Bhoir, Tanuja Naphade, Vaibhavi Shinde and Sameer Talekar in partial fulfilment of the requirements for the subject **Community Engagement Project** (SE, 1st Semester) of degree of Bachelor of Engineering in second Year Engineering from Dr. D. Y. Patil Institute of Technology, Pimpri during the academic year 2025-2026

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Acknowledgement

We take this opportunity to thank all those who have contributed in successful completion of this project work. I sincerely wish to express gratitude to our Project Guide Mrs. Chetana Shravage for full support, expert guidance, and encouragement and kind cooperation throughout the project work. I am greatly indebted to him/her for his help throughout project work. We are also very thankful to our Head of Department for their constant support and guidance.

Abstract

Communication is an essential part of life, and people with disabilities, especially those who are deaf or mute, often face difficulties in communicating with others. This project aims to develop a cost-effective, real-time hand gesture recognition system using flex sensors to assist people with hearing and speech impairments. A wearable glove embedded with five flex sensors (one on each finger) captures hand gestures based on finger bending patterns. These analog readings are processed using an Arduino Uno, and a moving average filter is applied to smooth sensor noise.

Each finger's bent or straight state is determined by comparing its averaged sensor value against predefined thresholds. The resulting boolean states (true for bent, false for straight) are then used to generate a 5-bit binary code, with F1 as the Least Significant Bit (LSB) and F5 as the Most Significant Bit (MSB). This binary combination uniquely identifies letters of the English alphabet, starting from 'A' for the condition (f1 && !f2 && !f3 && !f4 && !f5). This logic allows structured and deterministic recognition of individual letters from A to Z.

The detected letter is then displayed in real-time on a 16×2 LCD along with live sensor values for debugging and demonstration. The system is lightweight, easy to use, and highly customizable, with potential for future expansion such as voice output, word prediction, or wireless transmission. Compared to vision-based systems, the glove is less affected by lighting conditions and offers better portability and robustness.

In conclusion, the developed glove-based gesture recognition system provides a simple yet powerful tool for bridging the communication gap faced by the deaf and mute community, promoting inclusivity and accessibility through low-cost embedded technology.

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List of Acronyms

Acronym	Full Form / Description
AI	Artificial Intelligence
ADC	Analog-to-Digital Converter
ASL	American Sign Language
DIY	Do It Yourself
HCI	Human-Computer Interaction
IoT	Internet of Things
LCD	Liquid Crystal Display
UNO	Universal Numbering Organization
USB	Universal Serial Bus
VCC	Voltage Common Collector (<i>power supply voltage</i>)
Acronym	Full Form / Description
AI	Artificial Intelligence

Chapter I: Introduction

1) Domain

This project falls under the domain of Assistive Technology and Embedded Systems. Assistive technology refers to devices and systems specifically designed to enhance the functional capabilities of individuals with disabilities, empowering them to live more independent and inclusive lives.

In this project, we utilize basic electronics, sensor integration, and embedded microcontroller programming to help deaf and mute individuals communicate more easily with the general public. The system is based on a wearable glove embedded with flex sensors, an Arduino Uno microcontroller, and a 16×2 LCD display module. When the user performs specific hand gestures (similar to those in sign language), the system detects the bending of each finger using the flex sensors, processes the data in real-time, and displays the corresponding alphabet character on the LCD.

This solution bridges hardware and software in a compact, low-cost, and user-friendly form, illustrating a practical example of **human-computer interaction (HCI)** for inclusive communication.

2) Objectives

The main goal of this project is to design and develop a **smart glove** capable of recognizing basic sign language gestures and translating them into readable text.

The key objectives of the system are:

- To develop a wearable glove system integrated with flex sensors to detect finger bending and hand gestures.
- To process the analog sensor data using an Arduino Uno microcontroller, applying noise reduction through a moving average filter.
- To map gesture combinations into alphabetic characters using a binary-coded logic system, with F1 as the LSB and F5 as the MSB.
- To display the detected letters on a 16×2 LCD display for easy readability.
- To build an affordable, portable, and easy-to-use communication aid for deaf and mute individuals.

- To reduce dependency on interpreters or written communication by facilitating real-time, gesture-based interaction.
-

3) Problem Specification

Deaf and mute individuals rely primarily on sign language to convey their thoughts, needs, and emotions. However, sign language is not universally understood by the general population, leading to frequent miscommunication, frustration, and social exclusion. In many cases, communication depends on interpreters, written notes, or gestures — all of which are either slow, impractical in real-time, or unavailable in everyday interactions.

The core problem is:

How can we enable real-time, seamless communication between sign language users and non-signers using a simple, low-cost, and portable system?

This project proposes a solution in the form of a Sign Language Recognition Glove, which detects finger bending using flex sensors, processes the signals via Arduino Uno, and displays the corresponding alphabet letter on an LCD screen. Each gesture is identified using a binary logic approach, which significantly simplifies gesture recognition and increases speed and accuracy.

This system improves inclusivity, autonomy, and accessibility for individuals with hearing or speech impairments by providing a practical and wearable tool for daily communication in schools, workplaces, hospitals, and public interactions.

Chapter II: Literature Review

Gesture recognition using wearable devices has been extensively studied due to its applications in human-computer interaction, sign language translation, and assistive technologies. Several studies have explored the use of flex sensors, IMU sensors, and machine learning algorithms to interpret hand gestures.

Filipowska et al. [1] proposed a machine learning-based glove for gesture recognition, demonstrating that sensor-based gloves could accurately detect complex hand movements and translate them into text. Telluri et al. [2] also presented a low-cost gesture recognition glove, showing that flex sensors combined with either machine learning or simple binary mapping can reliably detect dynamic hand movements while remaining portable and affordable.

Choi et al. [3] focused on signal conditioning techniques, applying median filtering and moving average smoothing to reduce noise from sensor readings. This approach significantly improved the accuracy of detecting bent and straight finger positions reliably.

Sümbül et al. [4] developed a MEMS and flex sensor-based glove integrated with deep learning models, achieving high recognition rates for complex gestures. While accurate, the system complexity and cost were higher due to MEMS sensors and neural network processing.

Lakshmi et al. [5] implemented an Arduino and flex sensor-based glove for hand gesture recognition aimed at assisting speech-impaired individuals. Telluri et al. [2] similarly emphasized Arduino-based low-cost gloves, demonstrating practical, real-time gesture-to-text translation. While effective, these studies did not extensively explore advanced filtering or dynamic calibration for sensor variability.

El Rwelli et al. [6] focused on sign language recognition using convolutional neural networks. Although their approach achieved robust gesture recognition, it required substantial computational power and training data, limiting real-time deployment on simple microcontroller platforms.

In contrast, the proposed Smart Glove integrates dynamic calibration, median filtering, EMA smoothing, and binary mapping, enabling real-time, noise-resistant letter recognition with minimal hardware requirements. This approach combines the simplicity of binary mapping with robust signal conditioning, making it a practical and cost-effective solution for gesture-to-text translation in low-cost, portable systems.

Chapter III: Problem Identification

Communication is one of the most important aspects of human life. For deaf and mute individuals, sign language is the primary method of expressing thoughts and ideas. However, the majority of people do not understand sign language, which creates a major communication barrier in daily life, classrooms, workplaces, hospitals, and public places.

Traditional methods of communication, such as using interpreters or written notes, are often slow, inconvenient, and not always available, leading to misunderstandings and social isolation. Existing technological solutions, like camera-based gesture recognition systems, are often expensive, not portable, and require controlled lighting, while some sensor-based solutions are limited in accuracy or difficult to use.

The problem this project addresses is:

How to design a portable, cost-effective, and easy-to-use device that can detect hand gestures in real time and convert them into readable letters for people who do not understand sign language.

The goal is to bridge the communication gap between deaf/mute individuals and others, making social interaction faster, easier, and more inclusive.

Chapter IV: Methodology

❖ ALGORITHM USED

1. Read analog values from flex sensors F1 to F5.
2. Apply a moving average filter over **numReadings** samples to smooth the signal.
3. Compare averaged values to predetermined thresholds to classify each sensor as bent (true) or straight (false).
4. Form a 5-bit binary pattern: bit0 = F1, bit1 = F2, ..., bit4 = F5.
5. Match the binary pattern to a letter using **mapToLetter(...)**.
6. Display the sensor readings and the detected letter on the LCD.

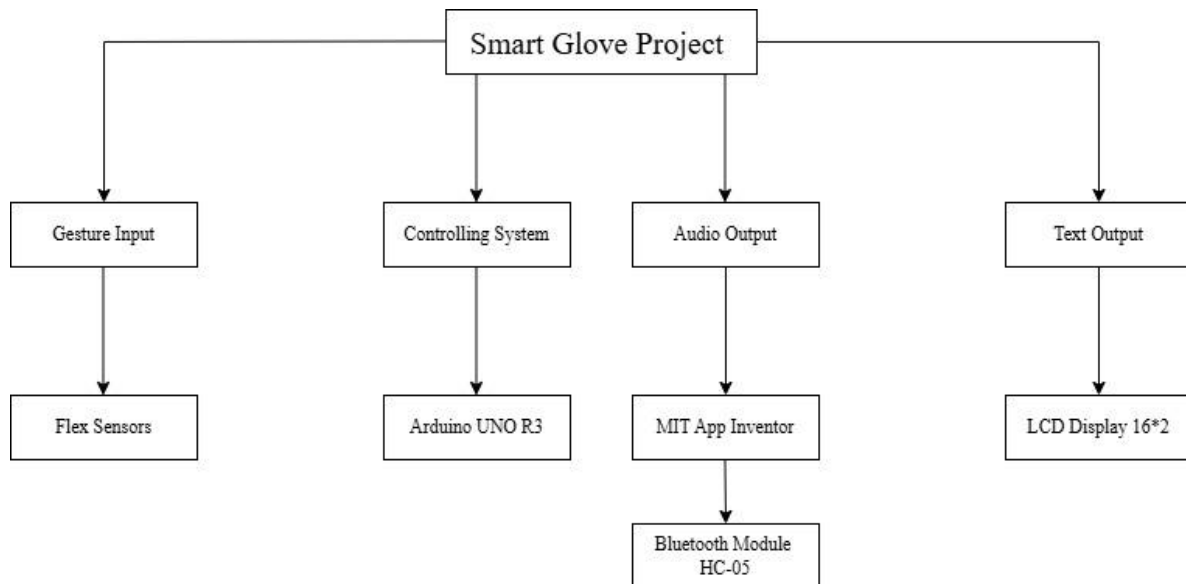


Fig 4.1: Flowchart

❖ SOFTWARE & HARDWARE REQUIREMENTS

1. Hardware:

- Arduino Uno
- 5 Flex sensors
- 16×2 LCD without I2C (LiquidCrystal interface)
- Jump wires, breadboard
- Glove substrate
- Power source (USB)

2. Software:

- Arduino IDE
- Tinkercad for simulation
- Serial Monitor

❖ ADVANTAGES

- Deterministic and easy to debug.
- Real-time display of gestures as text.
- Low cost, portable, lightweight.
- Easily extensible (e.g. more letters, phrases).

❖ LIMITATIONS

- Only supports preprogrammed letters.
- Calibration needed for different users (sensor thresholds may vary).
- Rapid gesture changes may be misread.
- Only single-letter display—no full-word construction in current form.

❖ APPLICATIONS

- Assistive communication for hearing-impaired individuals
- Educational tools in special schools

- Hospital and clinic interactions
- Public service counters for accessibility
- Future integration into IoT and smart systems

Chapter V: System Design

The Sign Language Glove system is designed to detect hand gestures and convert them into readable letters in real time. It follows a modular design consisting of input, processing, and output stages.

1) Input Module

- Each finger of the glove is fitted with a flex sensor.
- When a finger bends, the sensor's resistance changes, producing a varying analog voltage signal.
- The Arduino Uno reads these signals through its Analog-to-Digital Converter (ADC).

2) Processing Module

- The Arduino Uno acts as the central controller or brain of the system.
- Sensor readings are compared with predefined threshold values corresponding to specific gestures.
- The system uses a gesture recognition algorithm to map the detected gesture to the corresponding letter, word, or sentence.
- The algorithm also uses a moving average filter to reduce noise and improve accuracy.

3) Output Module

- Once a gesture is recognized, the corresponding text is displayed on an LCD display.
- The output is updated in real time, allowing users to communicate smoothly.
- The display can show individual letters or complete sentences, depending on the gestures performed.

4) System Architecture Diagram

The system can be visualized in three main blocks:

[Input] → [Processing] → [Output]

- Input: Flex sensors on glove fingers
- Processing: Arduino Uno microcontroller
- Output: LED/LCD display showing text

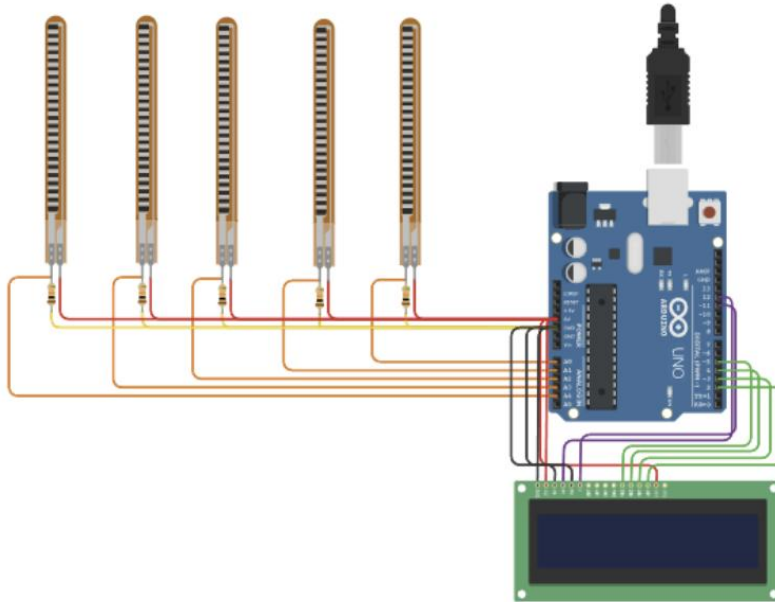


Figure 5.1: Tinkercad simulation of the flex sensor glove circuit

The circuit integrates five flex sensors, each attached to analog input pins A0 through A4 of the Arduino Uno. These sensors detect finger bending by varying resistance values. The Arduino processes these analog inputs, applies a moving average filter to reduce noise, and maps the sensor states to corresponding letters. The detected letters are then displayed on the 16×2 LCD, providing real-time feedback of the sign language gesture.

This simulation verifies the hardware connections and functionality before physical prototyping, ensuring that sensor readings and display outputs are correctly handled within the embedded system. A Block diagram of the circuit is shown in Fig. 5.2.

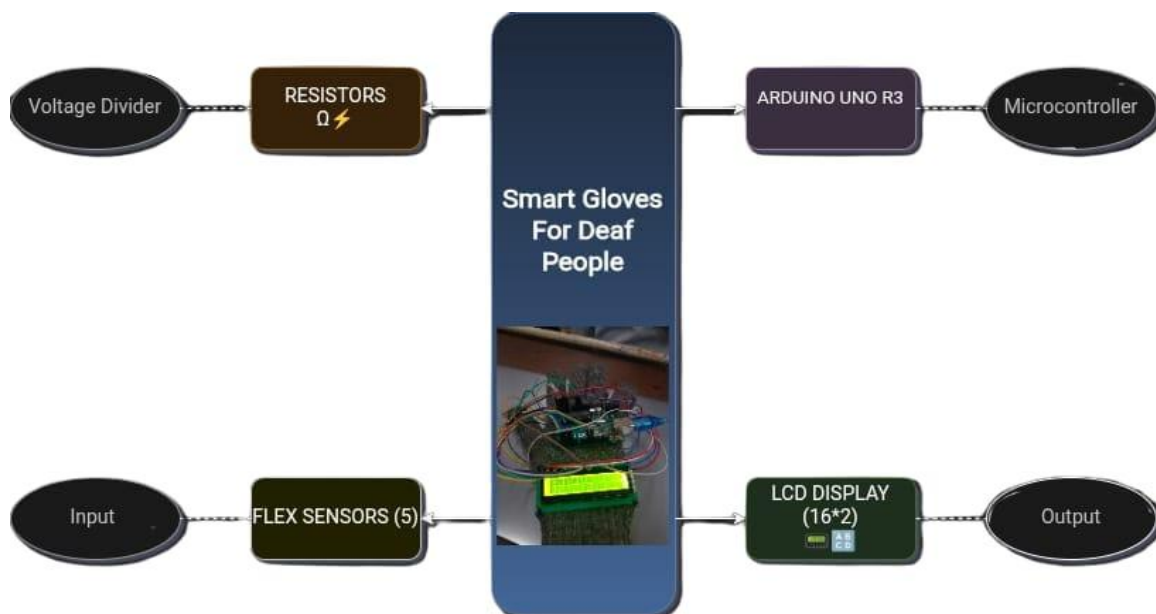


Figure 5.2: Block diagram of smart glove

5) Key Features

- Portable: Can be worn as a glove for daily use.
- Cost-effective: Uses basic electronic components.
- User-friendly: Easy to operate without specialized training.
- Real-time translation: Converts gestures into text instantly.

Flex state to Letter

Letter	F5 (MSB)	F4	F3	F2	F1 (LSB)	Binary Value
A	0	0	0	0	1	00001
B	0	0	0	1	0	00010
C	0	0	0	1	1	00011
D	0	0	1	0	0	00100
E	0	0	1	0	1	00101
F	0	0	1	1	0	00110
G	0	0	1	1	1	00111
H	0	1	0	0	0	01000
I	0	1	0	0	1	01001
J	0	1	0	1	0	01010
K	0	1	0	1	1	01011
L	0	1	1	0	0	01100
M	0	1	1	0	1	01101

N	0	1	1	1	0	01110
O	0	1	1	1	1	01111
P	1	0	0	0	0	10000
Q	1	0	0	0	1	10001
R	1	0	0	1	0	10010
S	1	0	0	1	1	10011
T	1	0	1	0	0	10100
U	1	0	1	0	1	10101
V	1	0	1	1	0	10110
W	1	0	1	1	1	10111
X	1	1	0	0	0	11000
Y	1	1	0	0	1	11001
Z	1	1	0	1	0	11010

Table 1: Binary Letters Mapping

Chapter VI: Implementation

Hardware Assembly

- Mount flex sensors along fingers of the glove.
- Wire sensors to analog pins A0–A4 via voltage divider circuits.
- Connect the LCD to digital pins 12,11,5,4,3,2.
- Supply power via USB or battery.

Software Implementation

- Initialize arrays for moving averages.
- In loop(), call updateAverage(...) for each sensor to get smoothed values.
- Determine boolean states f1...f5 by comparing with thresholds (e.g. flex1Avg < bentThresholdFlex1).
- Pass boolean states into mapToLetter(...) to get the detected letter.
- Display readings and letter on the LCD.
- Optionally, print to Serial Monitor every second.

Arduino IDE code:

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

const int flexPins[5] = {A0, A1, A2, A3, A4};

float ema[5];

int base[5];

int margin[5];

const float alpha = 0.4;

const int FIXED_MARGIN = 5;

const float NOISE_K = 1.0;

char stableLetter = '-';

char prevLetter = '-';

int sameCount = 0;

const int stableLimit = 2;

String sentence = ""; // store letters

int median3(int a, int b, int c) {
```

```

if ((a >= b && a <= c) || (a <= b && a >= c)) return a;
if ((b >= a && b <= c) || (b <= a && b >= c)) return b;
return c;
}

float getNoise(int pin) {
    int samples = 20;
    float sum = 0;
    int vals[20];
    for (int i = 0; i < samples; i++) {
        vals[i] = analogRead(pin);
        sum += vals[i];
        delay(5);
    }
    float mean = sum / samples;
    float sq = 0;
    for (int i = 0; i < samples; i++) sq += pow(vals[i] - mean, 2);
    return sqrt(sq / samples);
}

char mapToLetter(bool f1, bool f2, bool f3, bool f4, bool f5) {
    if (f1 && !f2 && !f3 && !f4 && !f5) return 'A';
    if (!f1 && f2 && !f3 && !f4 && !f5) return 'B';
    if (f1 && f2 && !f3 && !f4 && !f5) return 'C';
    if (!f1 && !f2 && f3 && !f4 && !f5) return 'D';
    if (f1 && !f2 && f3 && !f4 && !f5) return 'E';
    if (!f1 && f2 && f3 && !f4 && !f5) return 'F';
    if (f1 && f2 && f3 && !f4 && !f5) return 'G';
    if (!f1 && !f2 && !f3 && f4 && !f5) return 'H';
    if (f1 && !f2 && !f3 && f4 && !f5) return 'I';
    if (!f1 && f2 && !f3 && f4 && !f5) return 'J';
    if (f1 && f2 && !f3 && f4 && !f5) return 'K';
    if (!f1 && !f2 && f3 && f4 && !f5) return 'L';
    if (f1 && !f2 && f3 && f4 && !f5) return 'M';
    if (!f1 && f2 && f3 && f4 && !f5) return 'N';
    if (f1 && f2 && f3 && f4 && !f5) return 'O';
    if (!f1 && !f2 && !f3 && !f4 && f5) return 'P';
}

```

```

if (f1 && !f2 && !f3 && !f4 && f5) return 'Q';
if (!f1 && f2 && !f3 && !f4 && f5) return 'R';
if (f1 && f2 && !f3 && !f4 && f5) return 'S';
if (!f1 && !f2 && f3 && !f4 && f5) return 'T';
if (f1 && !f2 && f3 && !f4 && f5) return 'U';
if (!f1 && f2 && f3 && !f4 && f5) return 'V';
if (f1 && f2 && f3 && !f4 && f5) return 'W';
if (!f1 && !f2 && !f3 && f4 && f5) return 'X';
if (f1 && !f2 && !f3 && f4 && f5) return 'Y';
if (!f1 && f2 && !f3 && f4 && f5) return 'Z';
return '-';
}

void setup() {
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.print("Calibrating...");
  delay(2000);
  for (int i = 0; i < 5; i++) {
    base[i] = analogRead(flexPins[i]);
    margin[i] = FIXED_MARGIN + getNoise(flexPins[i]) * NOISE_K;
    ema[i] = base[i];
  }
  lcd.clear();
  lcd.print("Ready!");
  delay(1000);
  lcd.clear();
}

void loop() {
  bool bent[5];
  String binaryStr = "";
  for (int i = 0; i < 5; i++) {
    int val = median3(analogRead(flexPins[i]), analogRead(flexPins[i]), analogRead(flexPins[i]));
    ema[i] = alpha * val + (1 - alpha) * ema[i];
    bent[i] = abs(ema[i] - base[i]) > margin[i];
  }
}

```

```

// build 0/1 string for Serial Monitor
binaryStr += bent[i] ? "1" : "0";
}
// Print raw binary states
Serial.println(binaryStr);
// Detect letter
char detectedLetter = mapToLetter(bent[0], bent[1], bent[2], bent[3], bent[4]);
// Debounce
if (detectedLetter == prevLetter) {
    sameCount++;
    if (sameCount >= stableLimit && detectedLetter != stableLetter && detectedLetter != '-') {
        stableLetter = detectedLetter;
        sentence += stableLetter;
        if (sentence.length() > 32) sentence.remove(0, 1); // keep last 32 chars
    }
} else {
    sameCount = 1;
    prevLetter = detectedLetter;
}

// Display on LCD: current letter + sentence
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Letter: ");
lcd.print(stableLetter);
lcd.setCursor(0, 1);
lcd.print(sentence);
delay(100);
}

```

The following images showcase the glove during initial testing phases, demonstrating the system successfully detecting hand gestures and displaying the corresponding letters on the LCD. These early results highlight the effectiveness of the sensor integration and the responsiveness of the display system.

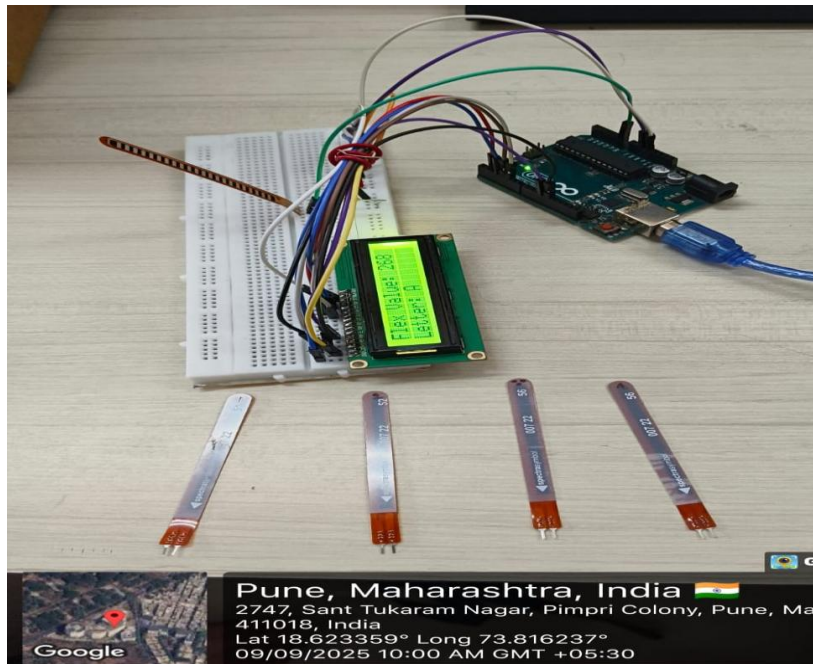


Figure 6.1: Glove wiring using a breadboard during initial testing.

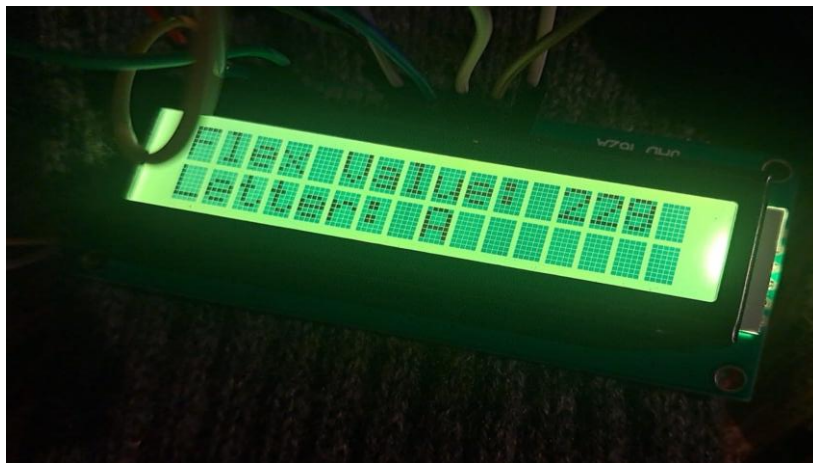


Figure 6.2: Displaying letter 'A' on the LCD during initial testing.

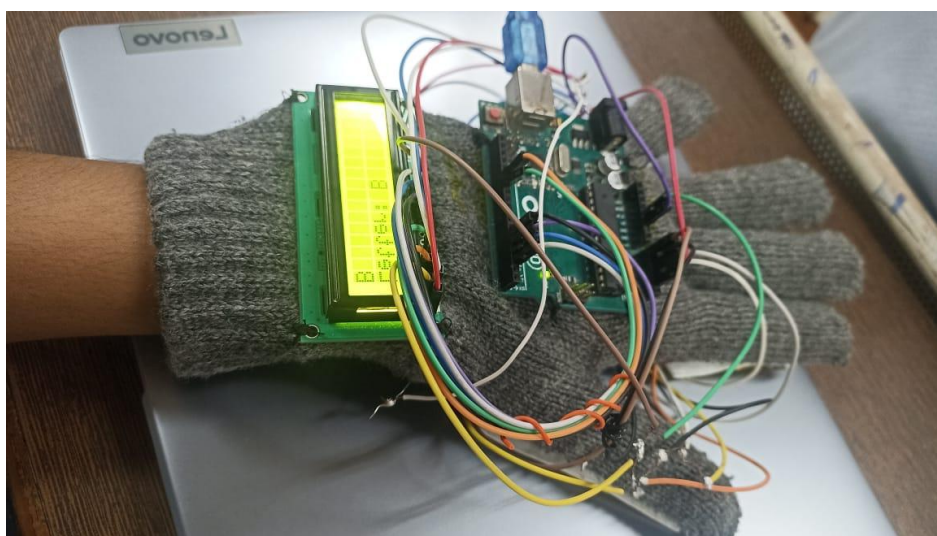


Figure 6.3: The Glove with wiring completed

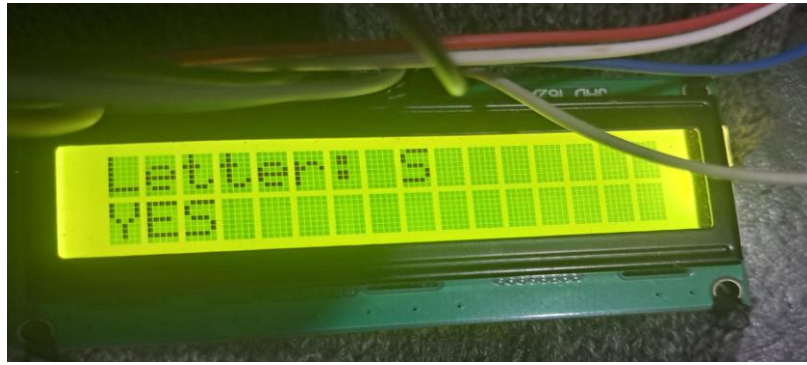


Figure 6.3: Displaying the word “YES” on the LCD

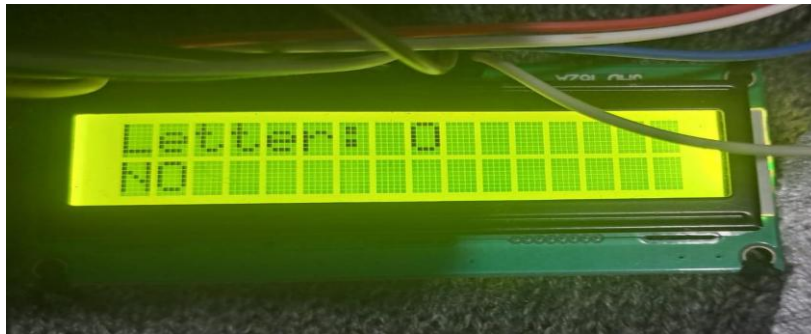


Figure 6.3: Displaying the word “NO” on the LCD



Figure 6.3: Displaying the word “OK” on the LCD

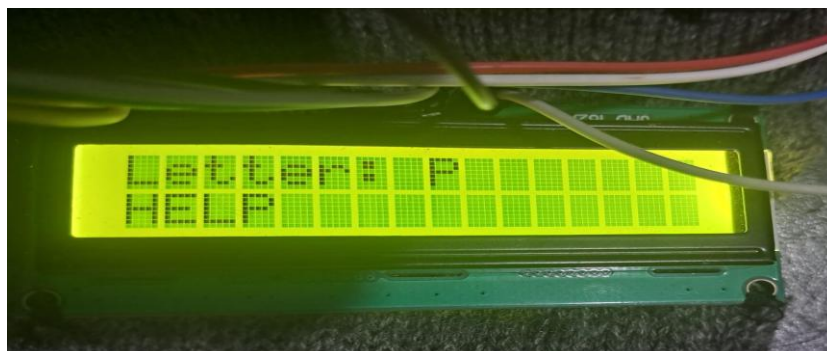


Figure 6.3: Displaying the word “HELP” on the LCD

Chapter VII: Conclusion and Future Scope

CONCLUSION

This project successfully demonstrates a wearable smart glove system that translates hand gestures into textual letters using flex sensors, Arduino Uno, and an LCD. The binary mapping scheme (F1 as LSB, F5 as MSB) allows a clear, deterministic logic for gesture-to-letter translation. The moving average filter and thresholding improve accuracy. The system displays both numeric sensor values and the detected letter on the LCD, making it useful in real-world communication. It is low cost, portable, and expandable.

FUTURE SCOPE

- Extend mapping for full words or phrases (not just single letters).
- Add a **speech synthesis module** to convert letters/words into spoken voice.
- Integrate **Bluetooth or Wi-Fi** to send detected letters to a smartphone or PC.
- Use **machine learning** to adapt thresholds and recognize more complex or variant gestures.
- Design a more ergonomic glove with flexible materials and better sensor alignment.
- Incorporate **gesture commands** to control IoT devices (lights, fans etc.).

Chapter VIII: References

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